



JFWM
2026
14-22 SEPT
POLYTECH
ANNECY-CHAMBERY
@ ANNECY, FRANCE



JAPANESE FRENCH WEEK ON MECHATRONICS
メカトロニクスの日本とフランス週間





メカトロニクスの日本とフランス週間

JFWM 2026

JAPANESE FRENCH WEEK ON MECHATRONICS

2026.09.14^{MON} - 22^{TUE}

@ ANNECY, FRANCE

Supports:



CLUB DES
ENTREPRISES
Université Savoie Mont Blanc



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7th edition of JFWM





JAPANESE FRENCH WEEK ON MECHATRONICS

Program

- Lectures
- Tutorials



Prof. Hideyuki Sawada

Waseda University, JP
School of Advanced Science and Engineering



Prof. Yuji Suzuki

Special Advisor to Dean, School of Engineering
The University of Tokyo, JP
Department of Mechanical Engineering



Dr. Takayuki Fujita

Associate Professor
University of Hyogo, JP
Graduate School of Engineering



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La Région
Auvergne-Rhône-Alpes
ENTREPRISES



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• One day conference : Mechatronics for Health

13h15 - Eloïse BISTON & Nicolas RABAH - ANTHOGR

Mechatronics in Dental Implantology : a cutting Edge Technology



13h55 - Hideyuki SAWADA - UNIVERSITY OF WASEDA

Applications of Mechatronics and AI in Health Research



14h35 - Alexandre QUARREY - AQ-TECH

Why most smart textile products never reach production, and what we learned by producing them



15h15 - Lukas LINDROTH - KINGS COLLEGE LONDON

Soft robotic needle guidance for ear interventions



15h55 - Rémi ROSSET-LANCHET - DEX SURGICAL

Robotic Arm for Minimally invasive Surgery





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Program

- NTN industrial site visit (Friday 18)





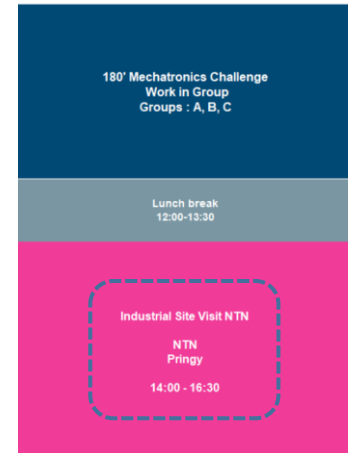
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Industrial site visit

- We will visit NTN on Friday afternoon

Friday Sept 18



Closed-toe shoes are **mandatory**



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Challenge

Grab it

Fabien Formosa – Professor, Polytech Anancy Chambéry

Luc Marechal – Assistant Professor, Polytech Anancy Chambéry

David Gibus – Assistant Professor, Polytech Anancy Chambéry

Blaise Girard – Mechanical Technician, Polytech Anancy Chambéry



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Program

- Mechatronics Challenge
- **6 TEAMS** of mixed of international students

FOUEJIEU NINZE	Jules	FISA5	A1	KIWAI		Tokyo univ	B1	MIURA	K	Tokyo univ	C1
BUISSSE	Charlotte	FISA5		GAUVRIT	Baptiste	FISA5		FOURNIER	Joseph	FISA5	
BERTEAUD	Elliot	FISA5		MOLLARD	Julien	FISA5		DOSSOU-YOVO	Samuel	FISA5	
MAILLET	Pierre Arthur	FISA5		ROLLIN	Arnaud	FISA5		FOUCHARD	Baptiste	FISA5	
MOHSIN	Syed Ali Ishaq	MAM2		CHESNAIS	Dorian	FISA5		ALIGROT	Mathieu	FISA5	
DALI	Karam	MAM2		YOUSSEF	Hussein	MAM2		YAZBICK	Mohamad Jaafar	MAM2	
				QIAO	Zhonghe	MAM2		ILYAS	Fizah	MAM2	
Hama	Mitsuo	Tokyo univ	A2	FURUKAWA	Hiroyasu	Tokyo univ	B2	KOJIMA	M	Tokyo univ	C2
ROUX	Anais	FISA5		MICHELIN	Alexis	FISA5		DESBRUS	Jérémie	FISA5	
BALPE LABASTROU	Antonin	FISA5		HOUDEAU	Alan	FISA5		FRELAT	Rémi	FISA5	
PAULIN	Louis	FISA5		BELORGEOT	Raphaël	FISA5		VANELSTRAETE	Nathan	FISA5	
COCHARD	Florian	FISA5		AL HOYEK	Elias	MAM2		COTRON	Hector	FISA5	
SHERRY	Karim	MAM2		SENE	Papa Madiera	MAM2		OUMA	Elsa	MAM2	
KAMDEM TCHEUKABA	Marc	MAM2						KHAN	Muhammad Adeel	MAM2	

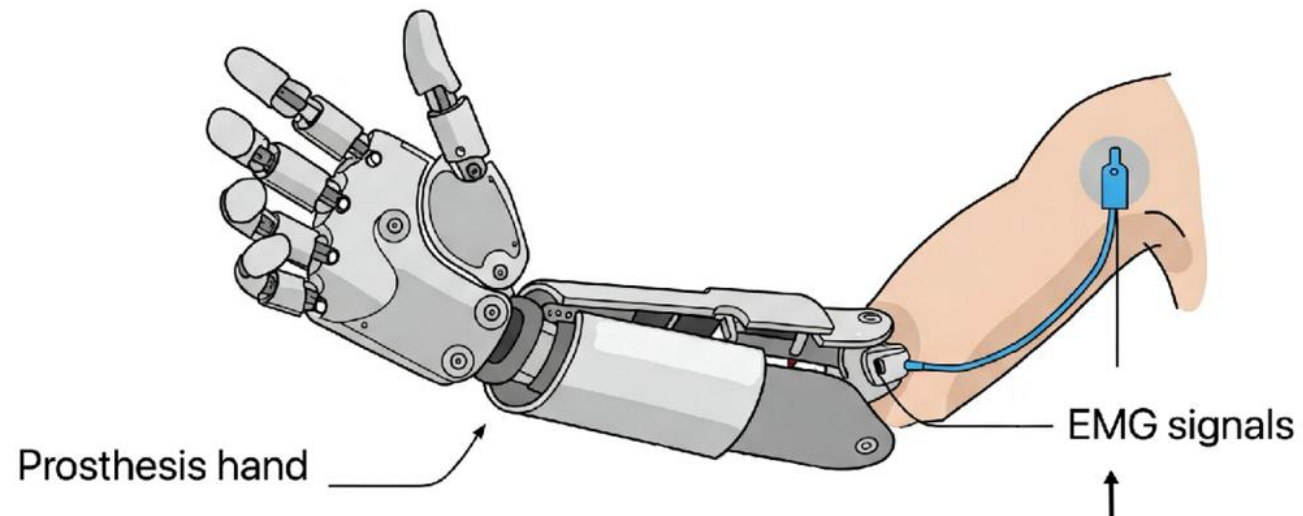


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Challenge brief

This year
challenge aims at
the design
and
implementation
of a Bionic
Gripper





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AIMS

- Design and create the gripper of the arm.
- Design and implement a transmission system for motors.
- Design and implement a pivot joint to mount the arm on the given frame.
- Develop Arduino code to control the arm and gripper
- Enable remote control of the system via a BioSignals (Rotation and Gripper actuation)
- The arm should be able to grab a soft object





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Groups

ALLOCATED group letters and students names

					A B C				
09:00									
09:15	Coffee 9:00 - Main Hall Polytech								
09:30									
09:45	Opening - 9:45-10:00 - A. Badel - B014								
10:00	Presentation: Program & "Challenge" - 10:00-10:15								
10:15	90' Lecture 10:15 - 11:45 Y. Suzuki, University of Tokyo Design of broadband vibration energy harvesting device B014				90' Lecture 10:00 - 11:30 T. Fujita, University of Hyogo, JP IoT modules & its tutorial by using SBC B014				
10:30									
10:45									
11:00									
11:15									
11:30									
11:45	Lunch break 11:45-13:15				Lunch break 11:30-13:15				
12:00									
13:15									
13:30									
13:45	Challenge: Start 13:15 - 14:45 Groups : A, B				90' Tutorial - 13:15 - 14:45 Y. Suzuki Design of broadband vibration energy harvesting device C114 - Group : C				
14:00					90' Tutorial - 13:15 - 14:45 Y. Suzuki Design of broadband vibration energy harvesting device C114 - Group : B				
14:15					90' Practical work 13:15 - 14:45 T. Fujita IoT modules tutorial using SBC C213 - Group : A				
14:30					Challenge 13:15 - 14:45 Group : C				
14:45									
15:00									
15:15	90' Tutorial - 15:00 - 16:30 Y. Suzuki Design of broadband vibration energy harvesting device C114 - Group : A				Challenge: Start 15:00 - 16:30 Groups : C, D				
15:30					Challenge 15:00 - 16:30 Groups : A, B, C				
15:45					90' Practical work 15:00 - 16:30 T. Fujita IoT modules tutorial using SBC C213 - Group : C				
16:00									
16:15									
16:30									



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Timeline

- 11 time slots of 90'
- Tuesday September, 22 Final demonstration

A jury will discuss and define which is the best accomplishment

(achievement of goals + demonstration + teamwork)



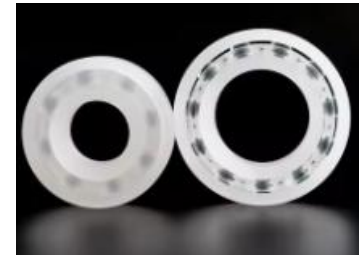


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Available material and tools

→ Mechanics and Electronics

- 1 frame
- 2 ball bearings
- 1 rotary sensor
- 1 Arduino board
- 1 Xiao BLE board
- 1 Stepper motor + pulley
- A set of timing belts
- Screws



→ Software and Fabrication

- Software: Solidworks, Simcenter 2404, Arduino IDE
- ABS 3D printers + Laser cutting machine





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Constraints



No glue ! No duct tape ! No cardboard ! No blu tack !



If you use it :





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Constraints

- All the 3D prints + cutting of parts will be done by Polytech
- Only 3 attempts to be qualified will be performed on Tuesday the 22nd
- Only one prototype per team

Expectations

1. The fastest and most precise positioning
(each of the prototypes will be confronted with an opponent to defined the best of all)
2. Use the less plastic material as possible (each prototype will be weighted)
3. Project management and task sharing. Teamwork.





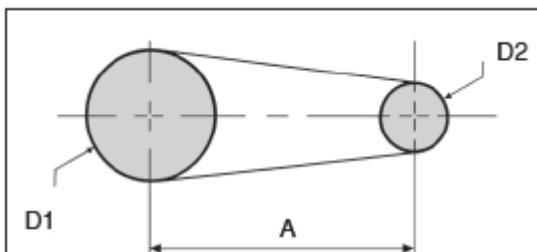
Technical data – timing belt

Détermination de la longueur de courroie

D1 = Diamètre primitif de la grande poulie
D2 = Diamètre primitif de la petite poulie
A = Entraxe
L = Longueur totale de la courroie
Z1 & Z2 = Nb dents
P = Pas

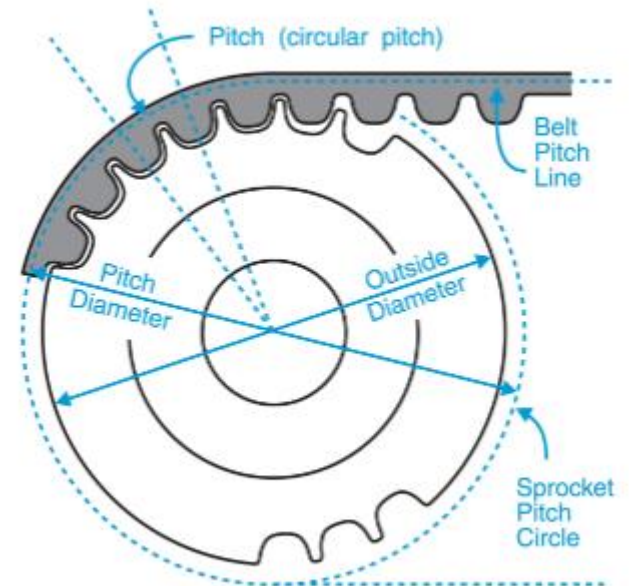
$$D1 = \frac{Z1 \cdot P}{\pi} \quad D2 = \frac{Z2 \cdot P}{\pi}$$

$$\text{Nb. de dents} = \frac{L}{P}$$

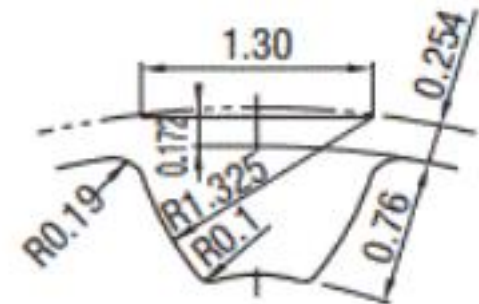


• Pour une transmission simple de deux poulies (réducteur ou multiplicateur) :

$$L \text{ (mm)} = \frac{\pi}{2} \cdot (D2 + D1) + 2 \cdot A + \frac{(D2 - D1)^2}{4 \cdot A}$$



Standard Tooth Profile



Tooth groove dimensions slightly change according to No. of teeth.

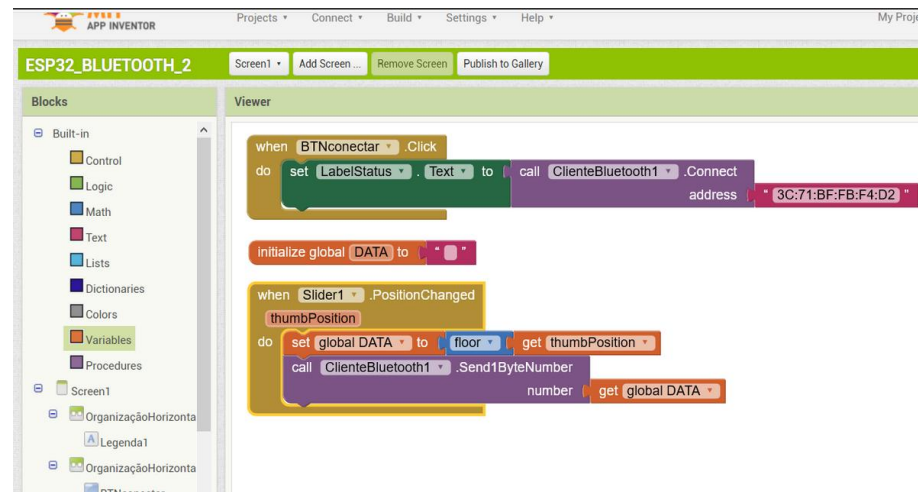
(Pitch: 2.0mm)



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Interface through cell phone via BLE

- Set up the Bluetooth communication between a cell phone and a Xiao BLE microcontroller
- Code an App with MIT APP Inventor2 or other
- Display the sensor measurement on a phone





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Resources on the Github repo

LucMarechal / JFWM2026

[Code](#) [Issues](#) [Pull requests](#) [Agents](#) [Actions](#) [Projects](#) [Wiki](#) [Security and quality](#) [Insights](#) [Settings](#)

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JFWM2026 - Challenge Student Groups.pdf	Add files via upload	21 minutes ago
JFWM2026_Program.pdf	Add files via upload	17 minutes ago
README.md	Update image in README.md	2 months ago

README

JFWM2026

About

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Contributors 1

LucMarechal Luc Marechal

JFWM2026





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Attendance is mandatory all the week

- Master and japanese students : sign on papers
- Apprentices :
 - sign on **NetyPareo** during lectures
 - sign on paper during practice

The screenshot shows a web interface for authentication. At the top, the word "Authentication" is displayed. Below it, there is a yellow silhouette of a person. To the right of the silhouette are two input fields: the first is labeled "Identifiant" and the second is for a password, indicated by dots and an eye icon. Below these fields is a green button labeled "Se connecter". At the bottom, there is a link that says "Mot de passe oublié ?".



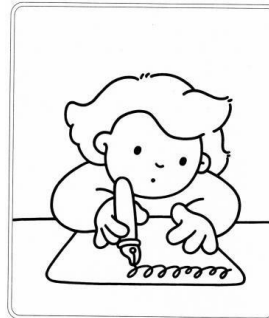
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Exam



- You will be evaluated by a final exam Tuesday 22nd
- It will be related to :
 - The Challenge
 - The lectures and tutorials
 - The Thursday conference and Friday industrial visit

**Take notes throughout
the week!**



Tuesday Sept 22

EXAM
9:00 - 10:00
B104

Mechatronics Challenge
Presentations, Jury deliberations & Awards
10:00 - 11:30
B104

JFWM 2024 Farewell Drink
11:30 - 13:30
Main Hall Polytech



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Day trip Saturday 19



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Day trip Saturday



- Hike at Hermitage Saint-Germain and Angon Waterfall about 2h walk.
- We will pick you up by car at 9:30 a.m. at the hotel
- Lunch at restaurant and return around 2:30 p.m. in Annecy
- French students are welcome !



- Infos will be sent by email



**Take hiking
shoes**



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Wifi and lunch

For Japanese students, you can access to wifi

- Connect to *usmb_visiteurs* and enter the codes

For lunch, you can go to restaurant of university and pay by credit card



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Have Fun !

Groups

GROUP A			TEAM	GROUP B			TEAM	GROUP C			TEAM
FOUEJIEU NINZE	Jules	FISA5	A1	KIWAI		Tokyo univ	B1	MIURA	K	Tokyo univ	C1
BUISSSE	Charlotte	FISA5		GAUVRIT	Baptiste	FISA5		FOURNIER	Joseph	FISA5	
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				QIAO	Zhonghe	MAM2		ILYAS	Fizah	MAM2	
Hama	Mitsuo	Tokyo univ	A2	FURUKAWA	Hiroyasu	Tokyo univ	B2	KOJIMA	M	Tokyo univ	C2
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PAULIN	Louis	FISA5		BELORGEOT	Raphaël	FISA5		VANELSTRAETE	Nathan	FISA5	
COCHARD	Florian	FISA5		AL HOYEK	Elias	MAM2		COTRON	Hector	FISA5	
SHERRY	Karim	MAM2		SENE	Papa Madiera	MAM2		OUMA	Elsa	MAM2	
KAMDEM TCHEUKABA	Marc	MAM2						KHAN	Muhammad Adeel	MAM2	